LIFELINE: Algorithm-selection report (proposal), Iteration 1. Group 18: Saneed Ullah (23i-2568), Hasnat Ahmed (23i-2652), Hussain Muhammad Ibrahim (23i-2626), AI-B				
Module	**Problem (n, m, V, E, W)**	**Candidates (time complexity)**	**Selection**	**Justification (property, assumption, failure case)**
A. Relief inventory	n indivisible crates with integer weights w in [1,20], stored values v in [10,100], capacity W. Maximise sum(v) with sum(w) <= W (0/1 knapsack).	Capacity DP O(nW) time, O(W) value memory (+ choices to rebuild); exhaustive O(n2^n); ratio-greedy O(n log n).	**Primary:** exact capacity DP + reconstruction. **Baseline:** ratio-greedy; brute force verifies S claims.	Small integer weights and seed-18 W <= 3,388 keep DP practical (max nW ~ 4.2m). Assumes integral kg. Large W makes reconstruction memory costly; trap 47 gives greedy only 55% of optimum. No day multiplier in data.
B. Route planning	G=(V,E), blocked wave roads, HQ 0, targets. Find reachable sites, fewest-road BFS path, and minimum-cost route using c(u,v)=minutes+penalty(v).	BFS O(V+E); binary-heap Dijkstra O((V+E) log V); array Dijkstra O(V^2); Bellman-Ford O(VE).	**Primary:** Dijkstra for cost; BFS for hops/reachability. **Baseline:** BFS-cost path and Bellman-Ford check.	E <= 3V and strictly positive step costs suit heap Dijkstra. Destination penalties make travel direction cost asymmetric. Must remove closed edges; return no route if disconnected. Negative edge weights would invalidate Dijkstra; trap 48 defeats fewest roads by >=1.5x. No storm multiplier.
C. Rescue scheduling	n windows [start, finish), values 1..100. Select compatible windows (touching allowed) to maximise count or total value.	Earliest-finish greedy O(n log n) (count); weighted interval DP O(n log n) (value); exhaustive O(n2^n); value-greedy O(n log n).	**Primary:** finish-time greedy / weighted DP with predecessor search. **Baseline:** value-greedy; brute on S.	Up to 3,000 windows and study-3 heavy overlap favour interval algorithms. Compatibility uses start >= previous finish; strict > is incorrect. Assumes one team, no travel or setup time; extra teams invalidate the model.
D. Emergency queue	m alarms (tick, priority 1..10, id). At each tick release at most one arrived alarm, highest priority first (tie: earlier tick/id).	Binary max-heap O(m log m); priority buckets O(m+10T); sorted-array insert O(m^2); scanned list O(m^2).	**Primary:** stable heap keyed (-priority,tick,id). **Baseline:** sorted array and bucket queue.	M/L/X arrival rate 1.2..1.8 per tick exceeds one service/tick; backlog can be large (1,826 in 18-47). Ten fixed priorities may make buckets faster. Priority aging needs a different update strategy.
E. Radio logs	Log length N, pattern P, reports lengths a,b, alphabet size sigma=38. Return all overlapping matches, minimum edits + script, and optimal prefix-free code.	KMP O(N+P) vs naive O(NP); edit DP O(ab) vs recursive exponential; Huffman O(N+sigma log sigma) vs fixed-length encoding O(N).	**Primary:** KMP, edit DP with backtrace, Huffman heap. **Baselines:** naive scan, recursive edits (small), fixed-length code.	Long logs (trap 49: 50k..100k) and frequent repeated prefixes punish naive search. Edit script needs O(ab) traceback storage. Huffman minimises expected bit length for observed symbol frequencies, assuming static, lossless binary coding.
F. Network planning	Undirected active roads carry integer cable costs. Connect all sites reachable from HQ with minimum total cable cost (MST of active component).	Kruskal+DSU O(E log E); heap Prim O(E log V); naive scan Prim O(V^2).	**Primary:** Kruskal+DSU. **Baseline:** Prim heap (same optimum).	Roads are sparse and cable weights positive (1..100). Exclude closed roads and isolate the HQ component before MST; do not confuse road travel minutes/penalties with cable cost. Disconnected graphs need a forest or explicit infeasibility.
G. Chains and cuts	n stage dimensions d0..dn: minimise matrix-chain scalar products. Drum prices p[1..L]: maximise revenue from exact cut lengths summing to L.	Matrix-chain DP O(n^3), O(n^2) space vs brute exponential; drum DP O(L^2) vs all cuts O(2^{L-1}).	**Primary:** interval DP + split reconstruction; rod-cut DP + cuts. **Baselines:** left-fold matrices and uncut drum.	Dimensions 2..100 and non-proportional prices mean greedy choices are unsafe. L reaches 2,000: quadratic drum DP is feasible; cubic chain DP may be slow for 300 stages. Requires compatible adjacent matrix dimensions.
H. Hard planning	Find satisfying 3-SAT safety assignment; exact ration subset target; shortest closed *Manhattan* tour; vertex cover of size <= k.	DPLL worst O(2^b); subset-sum DP O(nT); Held-Karp O($n^2 2^n$); k-cover branching O(2^k(n+E)).	**Primary:** DPLL, subset DP, exact S-tour/cover; heuristic tour & cover on large cases. **Baseline:** brute force (S), NN/greedy (L).	Planted feasibility gives checkable safety/rations/cover; it does not prove cover minimum. Exact TSP is feasible for S sites (5..9) but not 200..500 L sites. For larger instances heuristics lose optimality guarantees; report gaps explicitly.
I. Nearest sites and checksums	Find closest pair among n distinct (x,y) integer sites using squared Euclidean distance; multiply two decimal integers formed from 8-digit crate serials.	Closest pair divide-and-conquer O(n log n) vs brute O(n^2); Karatsuba O($d^{\log_2 3}$) vs schoolbook O(d^2).	**Primary:** closest-pair D&C and digit-array Karatsuba. **Baselines:** pair scan and schoolbook multiply.	Sites lie in $[0,999]^2$, up to 500, so brute checks correctness. Concatenated serials can yield thousands of digits, favouring Karatsuba. D&C assumes planar squared Euclidean distance; small-digit inputs favour schoolbook.

LIFELINE: Work division, Iteration 1. Group 18: Saneed Ullah (23i-2568), Hasnat Ahmed (23i-2652), Hussain Muhammad Ibrahim (23i-2626), AI-B. Every member can explain the whole submission.	
Member (name, roll number, section)	**Contribution in Iteration 1 and decision modules proposed for Iteration 2**
Saneed Ullah 23i-2568 AI-B	**Iteration 1 generator:** locations, roads, waves, sites, planted tour; trap 48 and variety-2 construction. **Validator:** V5 radio pattern/edit checks; V8 traps 46/49. **Exact claim:** tour_opt. **Approach:** story-to-numbers network/tour discussion. **Iteration 2 proposals:** B (routes), F (minimum cable network), I (closest pair and long-product).
Hasnat Ahmed 23i-2652 AI-B	**Iteration 1 generator:** crates/capacity, windows, incidents, pipeline/drum; traps 46/47 and variety-1 construction. **Validator:** V4 network; V6 planted tour; V7 signatures; V8 trap 48. **Exact claims:** cargo_opt, window_count_opt, window_value_opt, drum_opt. **Approach:** trap rationale. **Iteration 2 proposals:** A (inventory), C (rescue scheduling), G (matrix chains and drum cuts).
Hussain Muhammad Ibrahim 23i-2626 AI-B	**Iteration 1 generator:** cases/writer, radio, sensor cover, 3-SAT safety, rations; trap 49, variety-3/4 construction. **Validator:** loader/framework, V1–V3, V5 claim recheck, V6 cover/assignment/rations, V8 trap 47; broken tests/report. **Exact claim:** cover_min. **Approach:** planted objects/claims and the cross-scenario hard-decision summary. **Iteration 2 proposals:** D (queue), E (radio), H (hard planning).